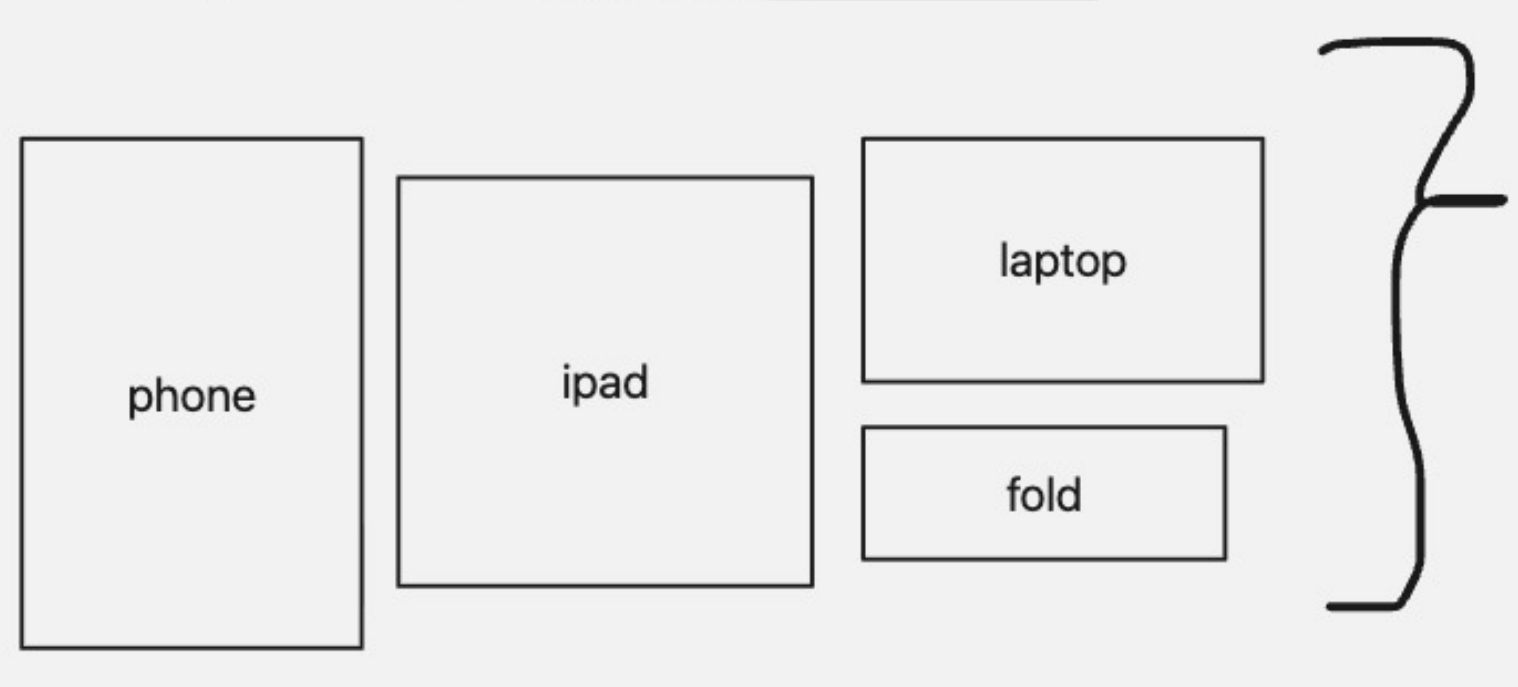
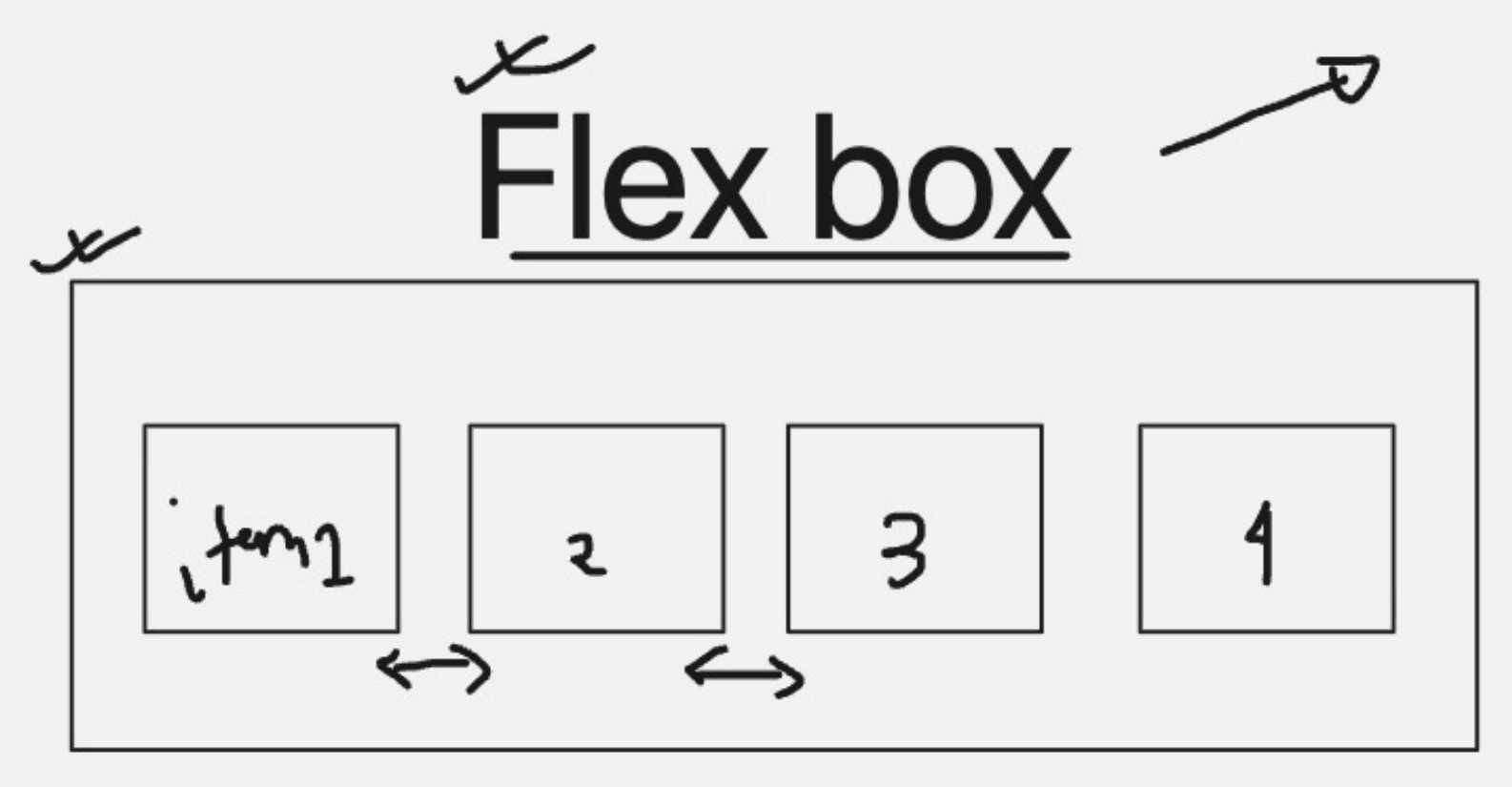
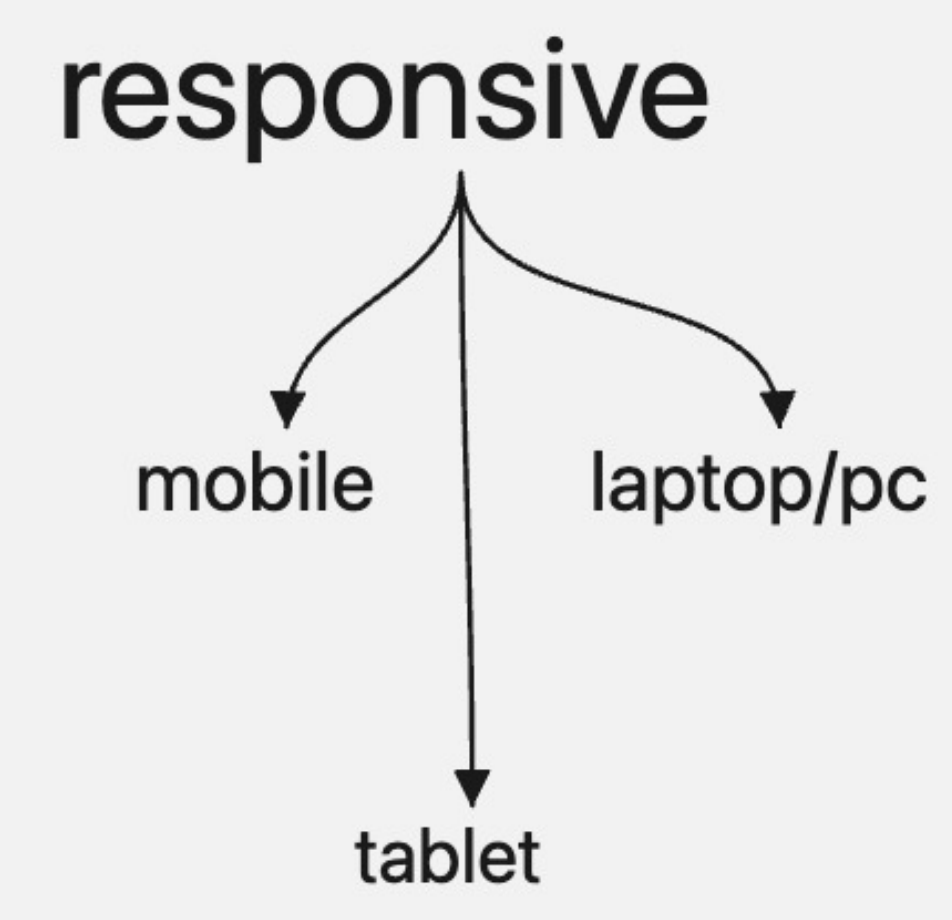
Agenda:

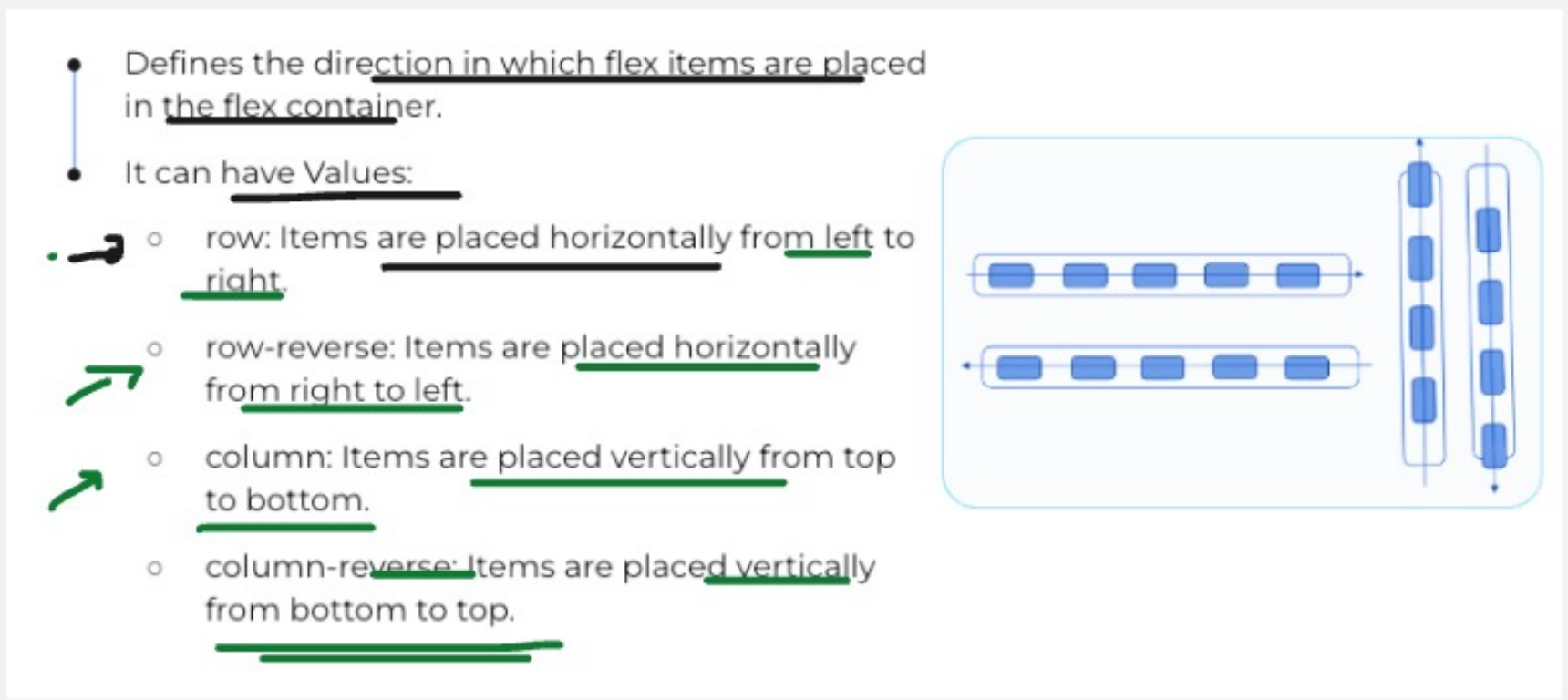
- recap
- flex box - demo
- Grid layout
- demo and task

Responsive UI

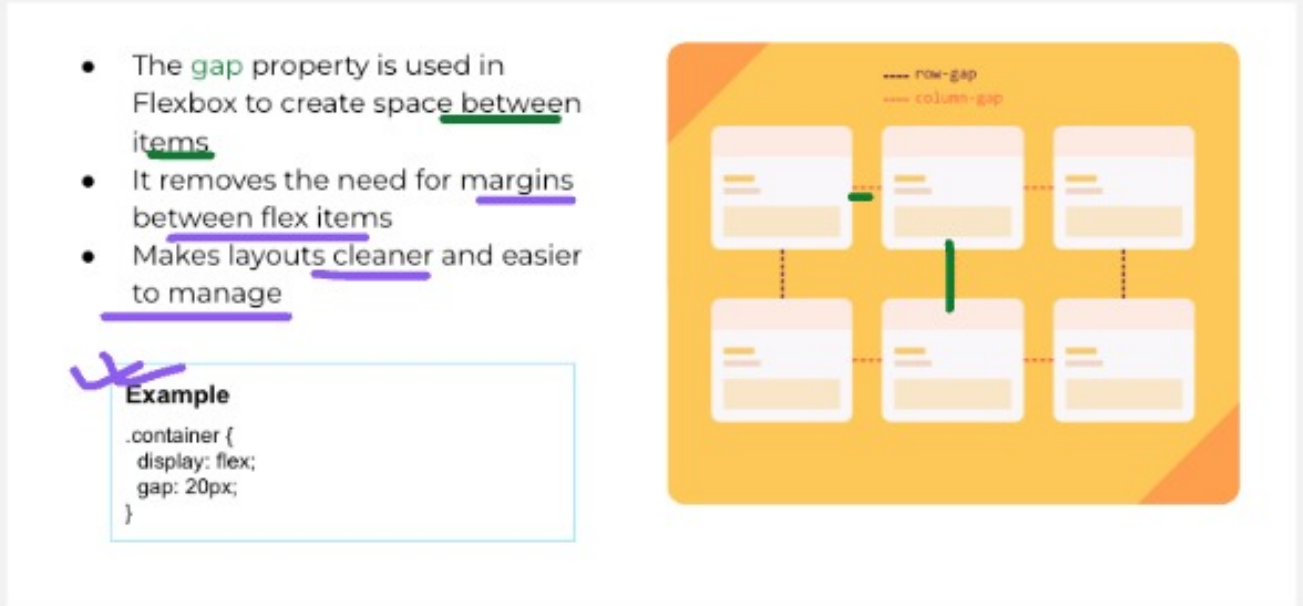








- Defines the direction in which flex items are placed in the flex container.
- It can have Values:
  - row: Items are placed horizontally from left to right.
  - row-reverse: Items are placed horizontally from right to left.
  - column: Items are placed vertically from top to bottom.
  - column-reverse: Items are placed vertically from bottom to top.



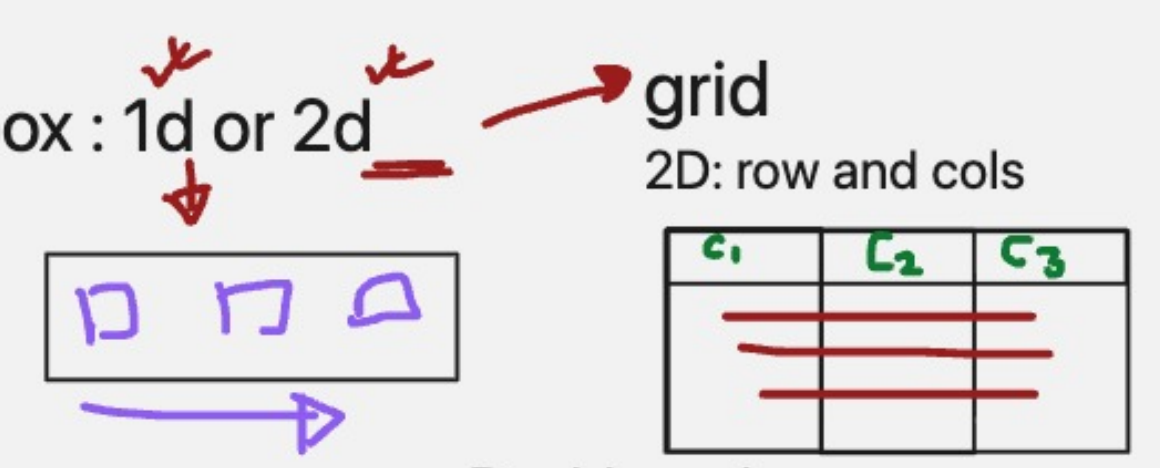
- The gap property is used in Flexbox to create space between items
- It removes the need for margins between flex items
- Makes layouts cleaner and easier to manage

Example

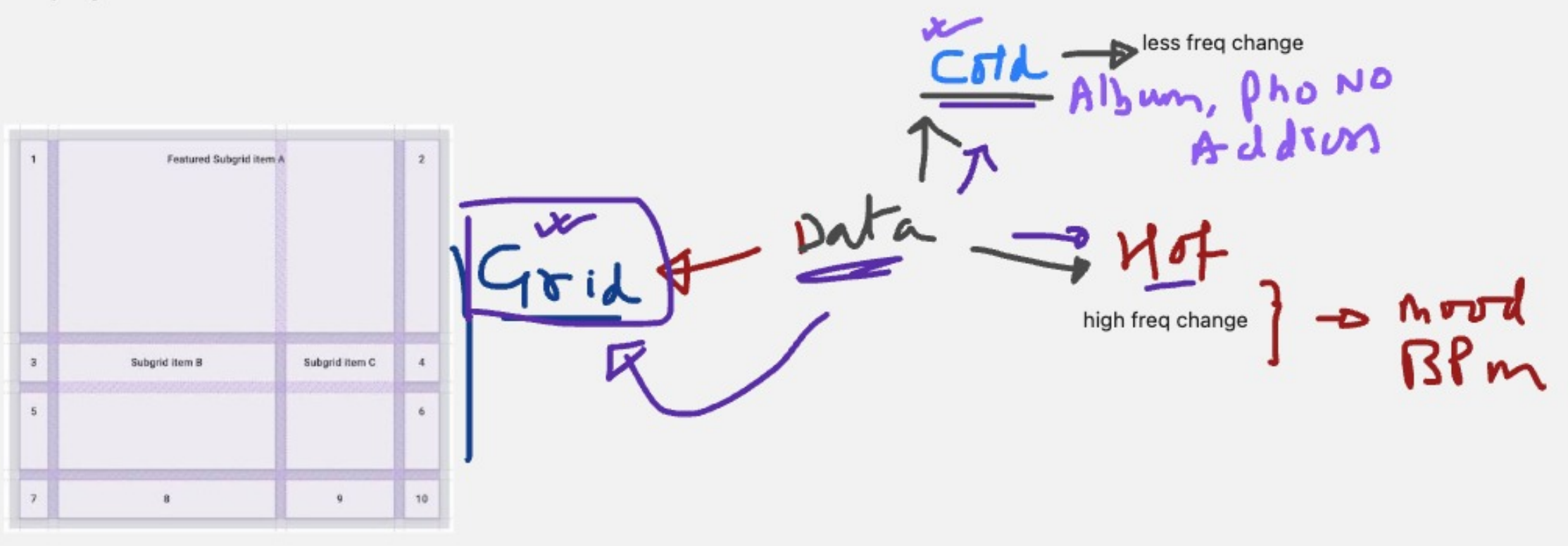
```
.container {
  display: flex;
  gap: 20px;
}
```

flex box : 1d or 2d → grid

2D: row and cols



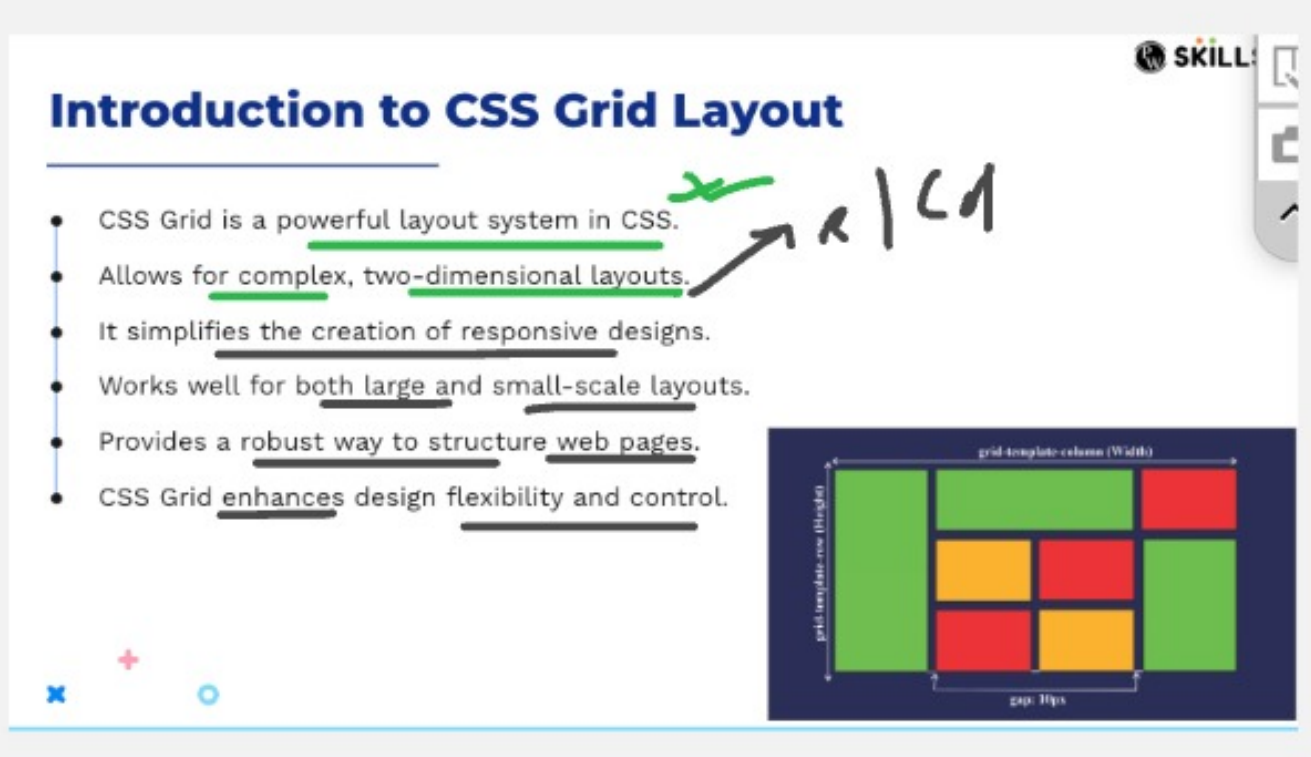
Dashboard:
view Dashboard:
employee of team delta



Cold data : system backup
HOT data : Stock

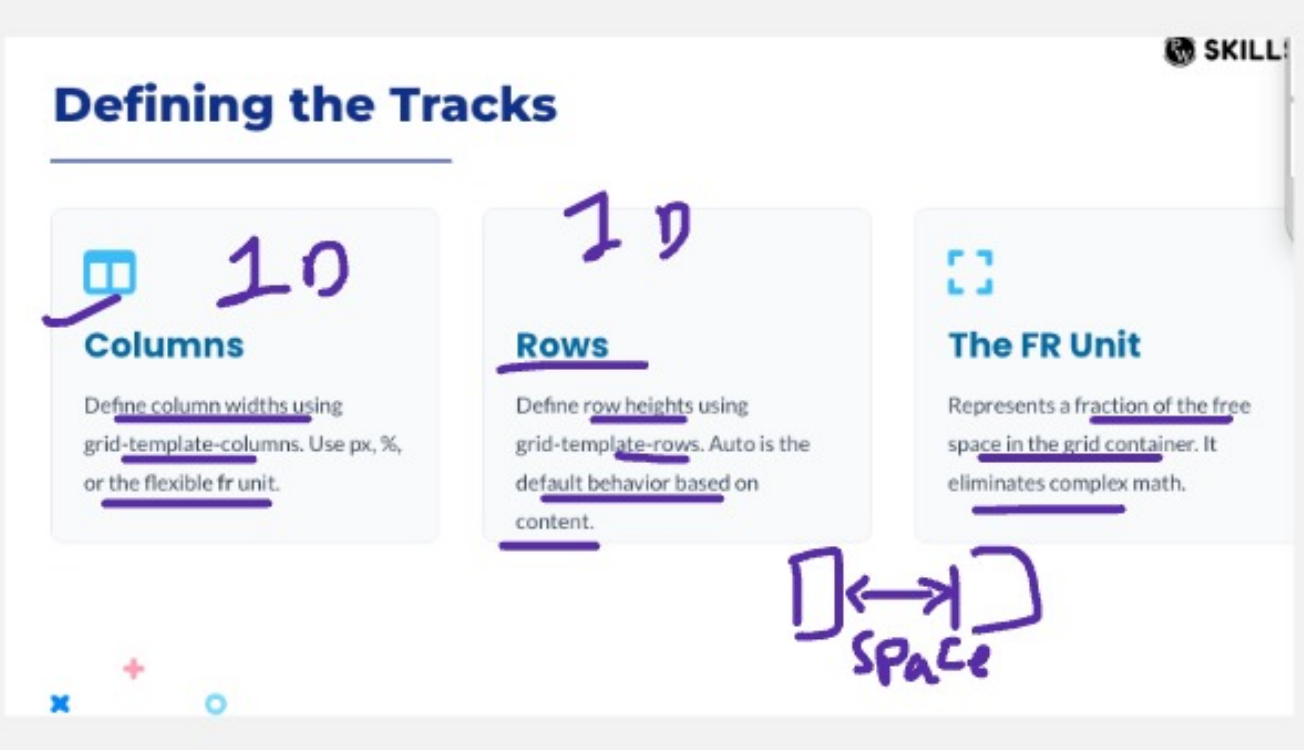
"Grid"

AG Grid



Introduction to CSS Grid Layout

- CSS Grid is a powerful layout system in CSS.
- Allows for complex, two-dimensional layouts. → R | Col
- It simplifies the creation of responsive designs.
- Works well for both large and small-scale layouts.
- Provides a robust way to structure web pages.
- CSS Grid enhances design flexibility and control.



Defining the Tracks

Columns 1D

Define column widths using grid-template-columns. Use px, %, or the flexible fr unit.

Rows 1D

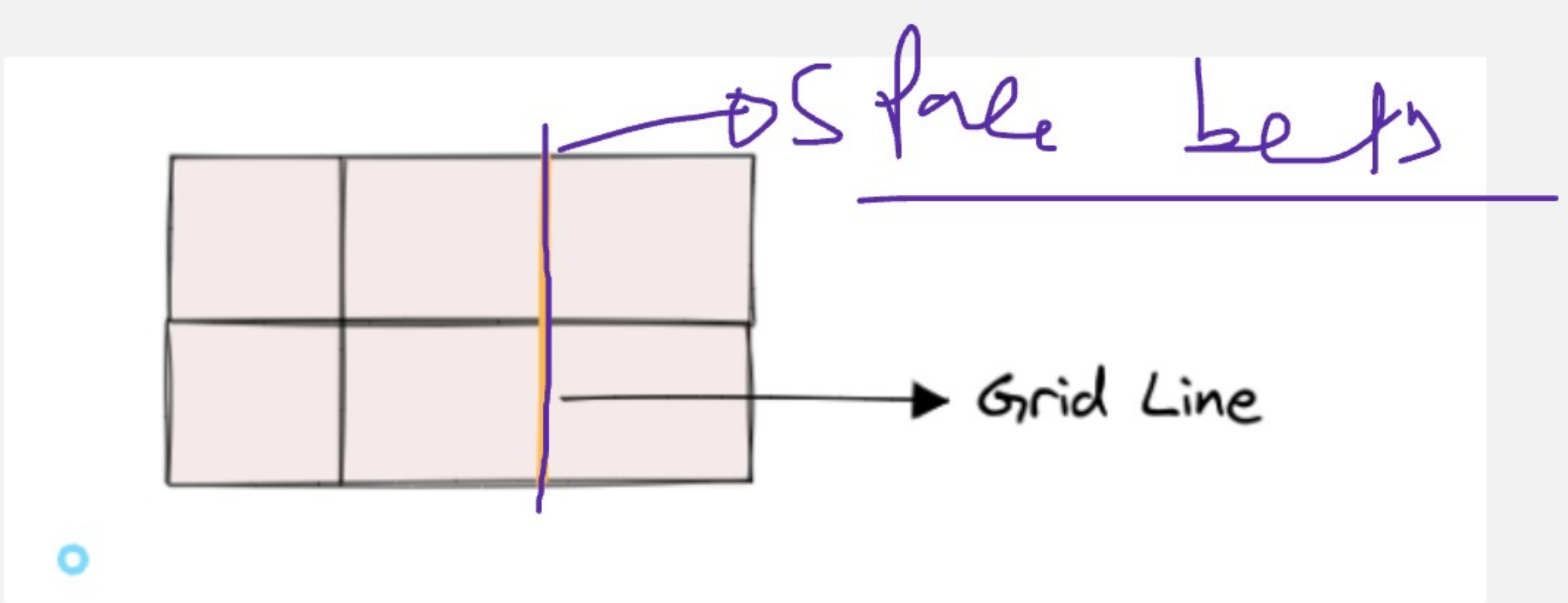
Define row heights using grid-template-rows. Auto is the default behavior based on content.

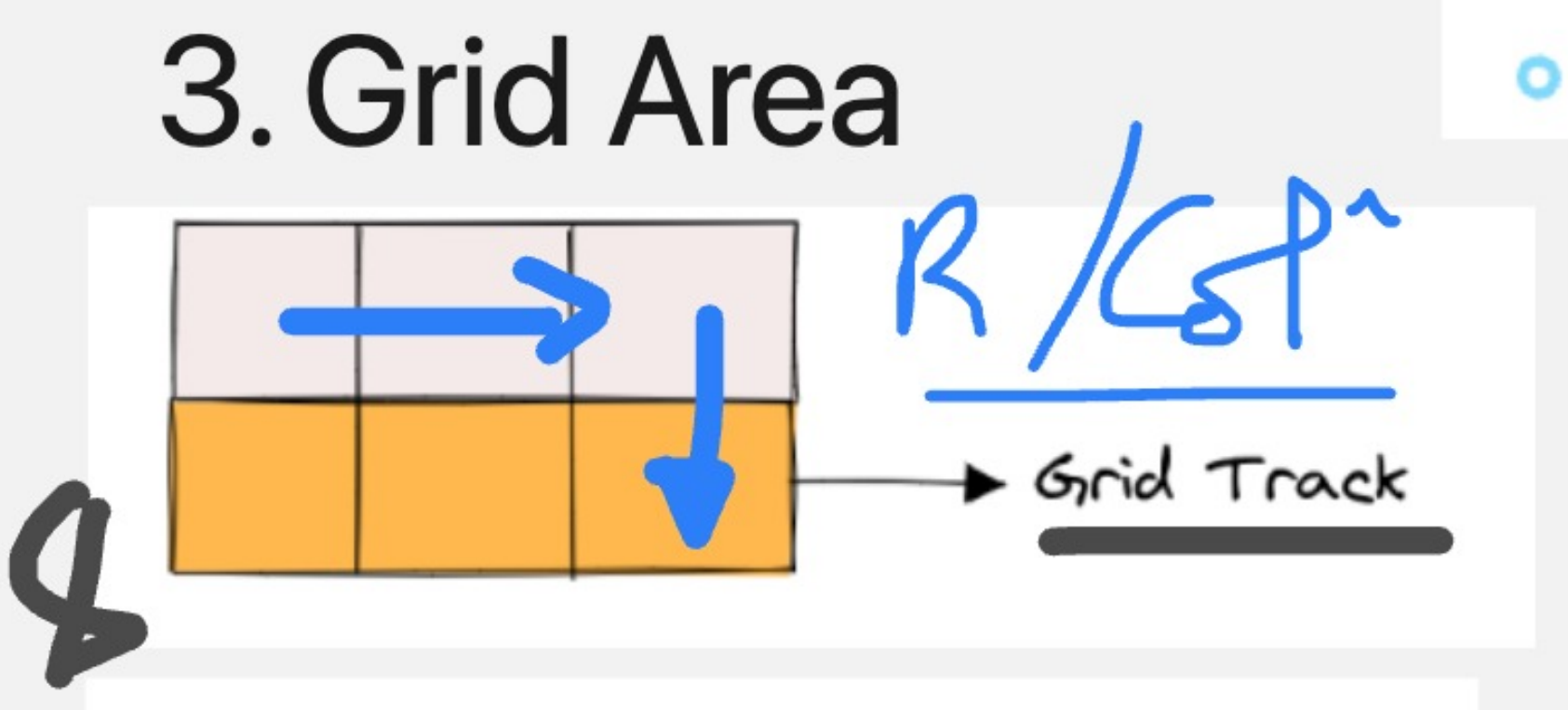
The FR Unit

Represents a fraction of the free space in the grid container. It eliminates complex math.

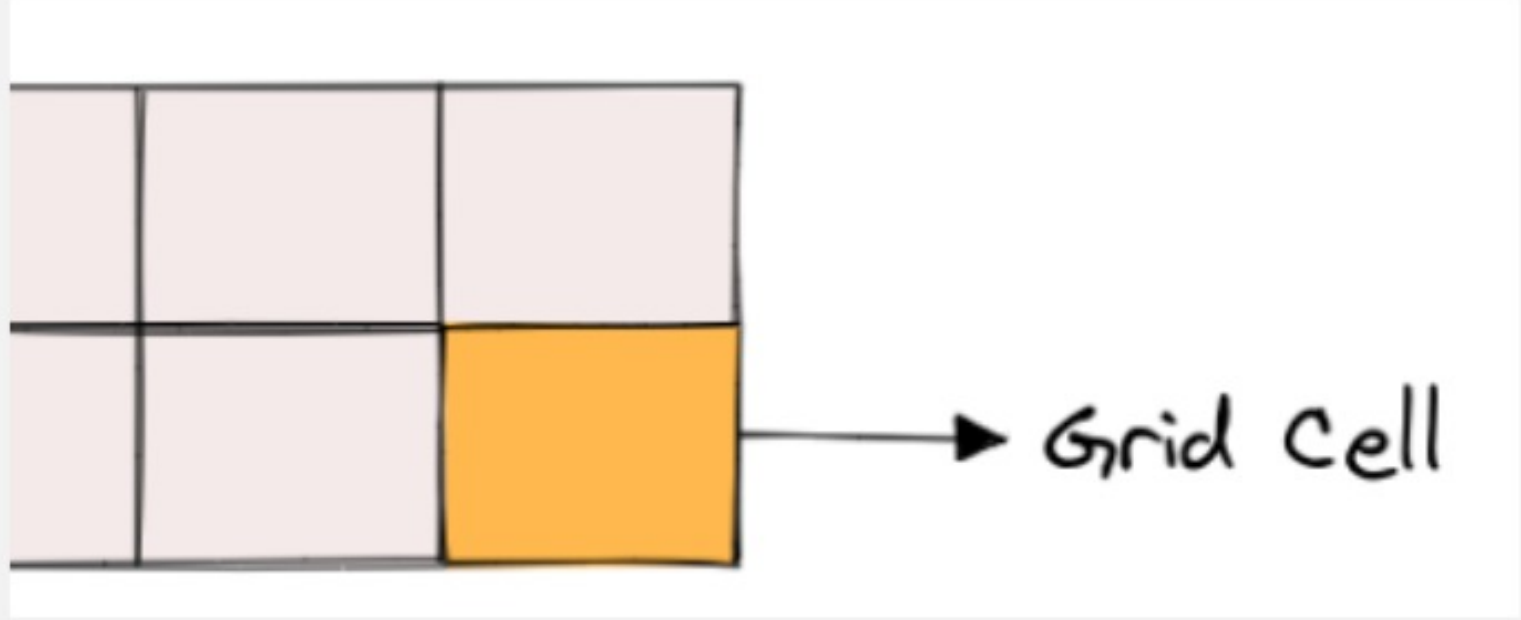
Grid properties:

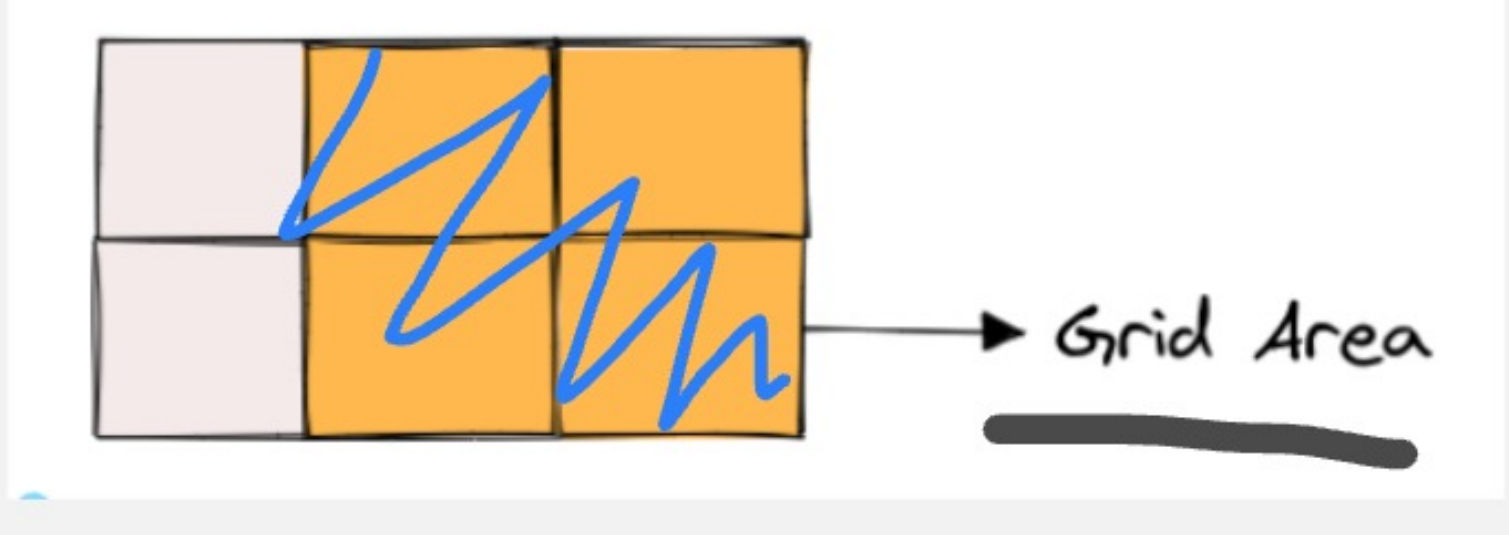
1. Grid Line
2. Grid cell
3. Grid Area

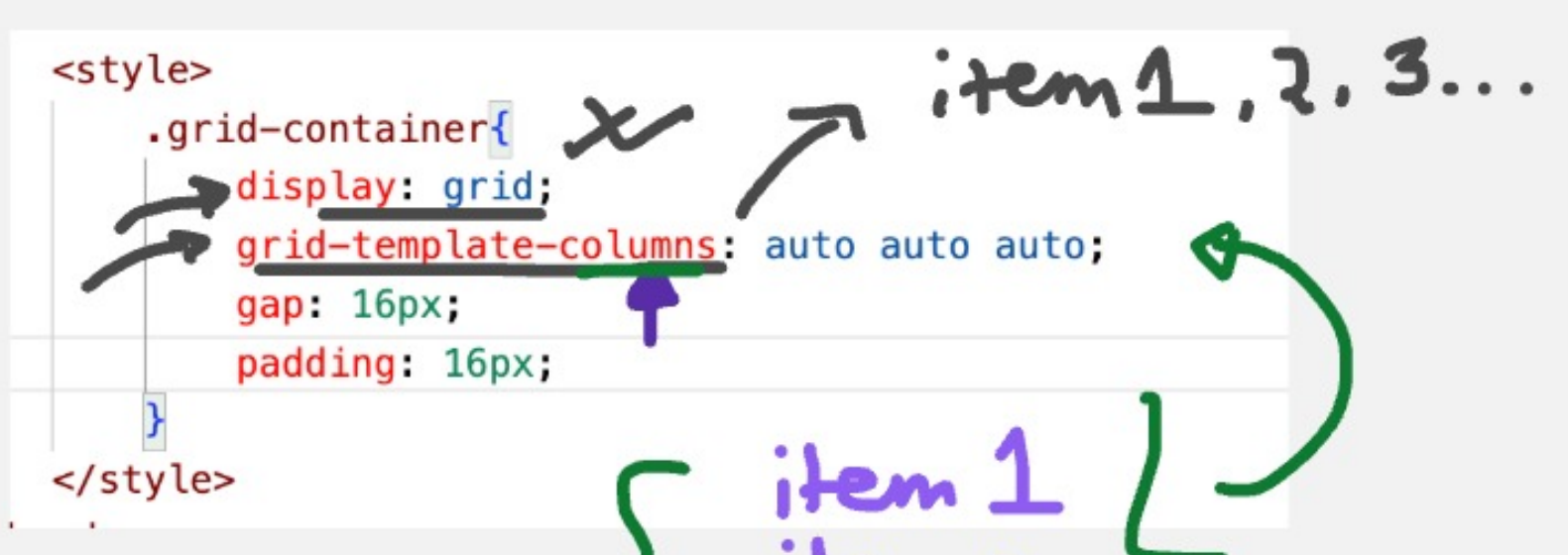




v/s







```
<style>
    .grid-container{
        display: grid;
        grid-template-columns: auto auto auto;
        gap: 16px;
        padding: 16px;
    }
</style>
```

item1, 2, 3...

ROW
COLS

item 1
item 2
item 3
item 4
5

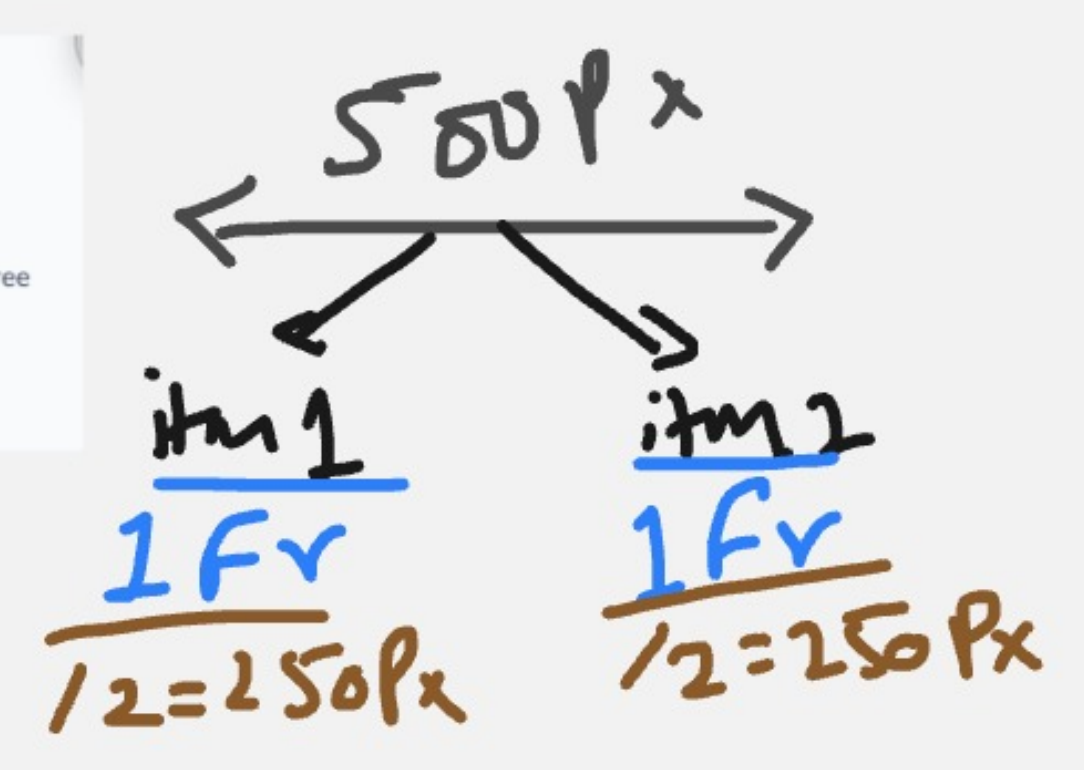
auto auto

Amount

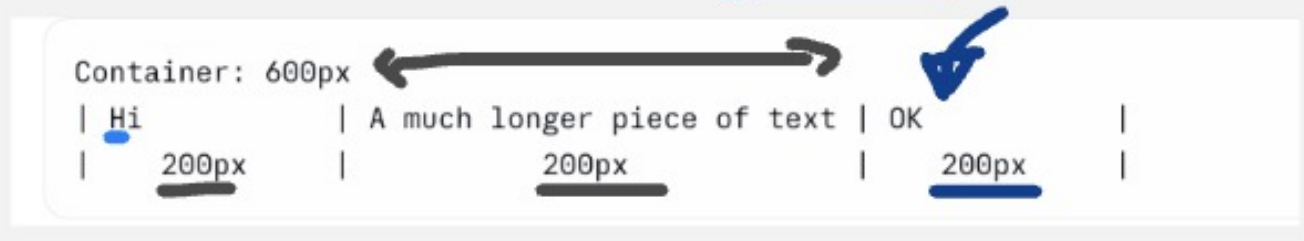
R, C
2, 3
4, 5

The FR Unit

Represents a fraction of the free space in the grid container. It eliminates complex math.

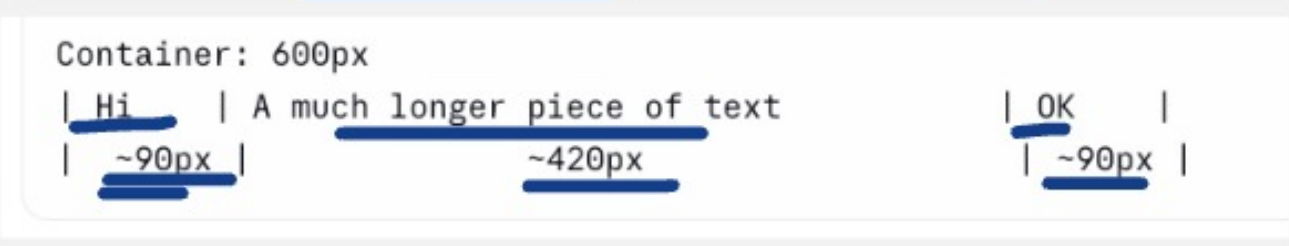


grid-template-columns: 1fr 1fr 1fr;



```
Container: 600px
| Hi              | A much longer piece of text | OK              |
|     200px       |            200px            |     200px       |
```

grid-template-columns: auto auto auto;



```
Container: 600px
| Hi    | A much longer piece of text                    | OK    |
|  ~90px |                  ~420px                       | ~90px |
```

auto-fit: when items should stretch to use all the space. This is the most common choice for cards and galleries.

→ auto-fill: when items should keep a consistent size even if there are only a few, like a product grid where one item shouldn't stretch across the whole page.